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AI-GENERATED · MICKY

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THE REFRAME

You asked which entry mode to choose; no financial case exists to evaluate any of them.

The brief assumes a build/acquire/partner decision is ready to be made. The analysis shows the IRR hypothesis is near-floor, no NPV model exists, blended margins are unquantified, and the TAM-to-SOM bridge has no bottom-up share-capture model. The question embeds the assumption that financial viability has been established and the only open issue is mode selection. That assumption is false. The real question is not which mode — it is whether the economics work at all under any mode. Mode selection cannot precede that answer.

RECOMMENDATION

Do not green-light any entry mode yet. The market is real. The financial case is not. Stop here until unit economics are modeled.

WEAKEST LINK

IRR and payback. No NPV model exists. No revenue ramp by customer segment. No blended margin by SKU mix. The only cost-side data is a $15–25M capex range from a Vertiv/Schneider/Eaton comparison blog — that is half an equation.

WHAT WOULD FLIP IT

Three things in sequence. One: a bottom-up P&L showing blended gross margin above 20% at $50M revenue, with BOM, channel rebates, and ramp costs explicit. Two: a feature-parity gap analysis against Vertiv Geist and Schneider APC with milestone dates — not a feature checklist. Three: confirmed approved-vendor-list status at two or more named hyperscalers or colos for the specific entry vehicle being considered.

FRAMEWORK RATIONALE

GE Market Attractiveness × Make-Buy-Ally

> **Simple go/no-go scorecard**
> A single score would average strong market evidence with missing economics and hide the binding diligence gaps.

> **Scenario planning**
> Scenario work is premature until ABB-specific unit economics and channel access are actually modeled.

> **Pure build-buy-partner matrix**
> Entry mode is a second-order choice until the market, access, economics, and roadmap gates clear.

BINDING CONSTRAINTS

1. **Unit economics** · Binding gate. No NPV, no margin model, no payback curve. Every other hypothesis is moot if this one fails. Confidence is near-floor — the lowest of any top-level hypothesis.
2. **Product competitiveness** · Second gate. Vertiv and Eaton already ship 17kW+ three-phase switched PDUs with deep remote monitoring. No technical gap assessment exists. Build path may be non-viable; buy/partner path is

unconfirmed.

3 **Roadmap resilience** · Material but not yet binding. Average rack density is only projected to reach 50kW by 2027, so the 100–200kW disruption risk is real but not imminent at scale. DC penetration data is absent, leaving the scenario open.

4 **Channel access** · Structurally problematic. ABB's channel runs through electrical contractors to facilities teams — a different buyer, process, and reseller ecosystem than IT procurement. Building a new sales motion typically takes 3–5 years, which breaks the 18-month coverage test.

5 **Market size** · The strongest hypothesis. Multiple independent sources place the rack PDU market at $2–3B in 2024 growing at 8–10% CAGR. Intelligent PDUs hold 61% share. The pond is large enough. The fish question is unanswered.

HYPOTHESIS DECOMPOSITION

Unit economics

Split into margin potential and IRR/payback. Strongest sub-hypothesis is margin potential directionally — intelligent PDUs sell at $750–$1,400 and deliver 20% energy savings — but no COGS, channel rebate, or mix data exists to confirm blended margin above 20%. Weakest sub-hypothesis is IRR and payback, which is near-floor: no NPV model, no revenue ramp, no payback curve. Gap-closing action: build a bottom-up P&L with BOM, channel discounts, and a DCF table at 15% WACC.

> METHODS NOT USED
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> **Payback-only screen** - Payback misses mode-specific capex, acquisition premium, and margin timing that determine whether the hurdle rate clears.
>
> **Incumbent gross-margin proxy** - Competitor margin ranges cannot substitute for ABB-specific BOM, channel leakage, and ramp costs.

Product competitiveness

Split into capability gap and brand permission. Strongest sub-hypothesis is brand permission — ABB is listed as a top-5 rack PDU supplier in Europe and holds a 26% regional share — but no approved-vendor-list confirmation at named hyperscalers or colos exists. Weakest sub-hypothesis is capability gap: Vertiv ships 17.2kW IP67 switched PDUs and Eaton ships 80kW rack cooling, with active roadmaps that may widen the gap before ABB closes it. Gap-closing action: commission a structured feature-parity teardown against Vertiv Geist and Schneider APC with explicit milestone dates.

> METHODS NOT USED
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> **Feature checklist as proof of competitiveness** - A feature list can show parity on paper but not whether ABB can close firmware, monitoring, and cost gaps on time.

> SCOPE CHOICES
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> **Build-only product assessment** - The decision explicitly includes buy and partner paths, which may inherit a competitive stack rather than build one.

Roadmap resilience

Split into density migration and DC distribution risk. Strongest sub-hypothesis is DC distribution risk — no strong contradicting evidence exists and 48V architecture remains dominant — but the 15% penetration threshold has no quantitative data behind it. Weakest sub-hypothesis is density migration: average densities are projected at only 50kW by 2027, well below the 100–200kW falsifier band, but no third-party density-band penetration forecast by segment exists. Gap-closing action: obtain an Uptime Institute or IDC report segmenting installed racks by power-density band with a 24-month forward forecast.

> METHODS NOT USED
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> **Single HVDC adoption forecast** - Available evidence does not support one confident penetration curve, so DC distribution is treated as a scenario risk.

> SCOPE CHOICES
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> **Hyperscale-only roadmap** - The accessible opportunity includes colocation and enterprise buyers whose density timelines may differ from hyperscalers.

Channel access

Split into existing channel reach and AVL via buy/partner. Strongest sub-hypothesis is existing channel reach — evidence consistently and credibly shows ABB routes through electrical distributors, not IT procurement — which actually argues against the parent claim. Weakest sub-hypothesis is AVL via buy/partner: one strong finding lists AWS and NTT Global as ABB customers, but no approved-vendor-list document confirms this for rack PDU specifically at three or more hyperscalers. Gap-closing action: obtain supplier portal or RFP award records from AWS, Equinix, or Digital Realty naming the target entity.

> CUT HYPOTHESES
>
> **Electrical channel can carry IT rack-PDU sales** - Facility-buyer relationships do not prove access to IT procurement or hyperscaler approved-vendor-list processes.

> METHODS NOT USED
>
> **Distributor-count proxy** - Counting electrical distributors would not establish coverage of named data-center IT decision-makers.

Market size

Split into market growth, intelligent mix, and TAM-to-SOM bridge. Strongest sub-hypothesis is intelligent mix — 61.42% intelligent PDU share in 2025 is well above the 30% falsifier threshold and corroborated by multiple independent sources. Weakest sub-hypothesis is TAM-to-SOM bridge: the case brief itself flags that China's $500M market yields only $75M accessible, and no bottom-up share-capture model exists to bridge from TAM to $50M SOM. Gap-closing action: build a geography-filtered SAM with explicit share-capture assumptions benchmarked against new-entrant win rates.

> METHODS NOT USED
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> **Top-down TAM share only** - Headline TAM share would not test whether accessible geographies can produce the lower revenue threshold.

> SCOPE CHOICES
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> **China-led revenue pool** - China may be large but is not reliably accessible to ABB for the target entry question, so it cannot carry the base SOM case.

JUDGMENT CALLS

IRR confidence floor

The only cost-side data is a $15–25M capex range from a 2024 Vertiv/Schneider/Eaton comparison blog post — not an ABB-specific investment model. No revenue ramp, no WACC-discounted cash flow, no payback curve exists anywhere in the evidence set.

> I rated this near-floor rather than outright false because missing evidence is not a triggered falsifier. The market is large enough that a viable financial case could exist — it simply has not been built. A bottom-up DCF with year-1 to year-5 contracted revenue by segment and an explicit capex schedule would resolve this.

Brand permission vs. approved-vendor-list status

ABB is listed as a top-5 rack PDU supplier in Europe and named alongside AWS and NTT Global as customers, but the sourcing document is a third-party market report, not a procurement record. No approved-vendor-list entry, qualified supplier register, or RFP award document is in evidence.

> I kept brand permission as weak rather than credible because market-report mentions do not equal formal procurement approval. The distinction matters: IT procurement at hyperscalers runs separate approved-vendor-list processes from electrical infrastructure contracts. Supplier portal confirmation from Equinix, AWS, or Digital Realty would resolve it.

DC distribution disruption timeline

The 15% penetration threshold has no quantitative backing. The Bloom Energy 2026 survey shows 45% of respondents expect DC architectures by end-2028 — that is intent, not installed base, and the survey is from an interested party.

> I held this at mixed rather than weak because average rack density is projected at only 50kW by 2027, which is well below the threshold where AC-to-DC conversion becomes structurally necessary at scale. A third-party density-band penetration forecast from Uptime Institute or IDC would settle the timeline question.

HOW I'D PUSH BACK

> The revenue target is market-size theater. Vertiv and Schneider collectively hold over 60% of the intelligent PDU segment and have decade-long approved-vendor relationships with every major hyperscaler and colo in Latin America and Southeast Asia. ABB has presented no win rate, no pilot deal size, and no existing customer pipeline that extrapolates to $50M. The pond is described but the hook is not.

> ABB's channel is structurally misaligned. Its distribution runs through electrical contractors to facilities and operations teams. Rack PDU purchasing at hyperscalers goes through IT procurement, separate reseller ecosystems, and approved-vendor-list processes that ABB has not demonstrated it is on. Building that sales motion from scratch typically takes 3–5 years — longer than the 18-month coverage test requires.

> The financial case does not exist yet. There is no blended margin model, no BOM, no channel rebate structure, and no DCF. Asking the team to choose between build, acquire, and partner before any of those inputs are available is asking them to allocate capital against an unresolved falsifier. The mode decision should be deferred until a bottom-up P&L clears the 15% IRR hurdle under at least one entry scenario.

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MEMO

ABB rack PDU

Attempt 1

delete this attempt